Jewel Lens — Stump the Model, Problem 2

Image-based jewelry catalogue matcher: upload a photo, get the top-5 closest catalogue matches with confidence scores, per detected item. Full design rationale and trade-offs are in **DECISIONS.md** — this document is the submission-facing writeup: what was built, how it performs, and where it breaks.

Part A — Catalogue Matcher

Catalogue: ~3,800 listings / ~13,300 images, scraped from public Shopify `/products.json` endpoints across 8 stores, spanning 3 categories (earring, necklace, bracelet) and a 16-colour palette, cleaned via perceptual-hash dedup + CLIP zero-shot label QA.

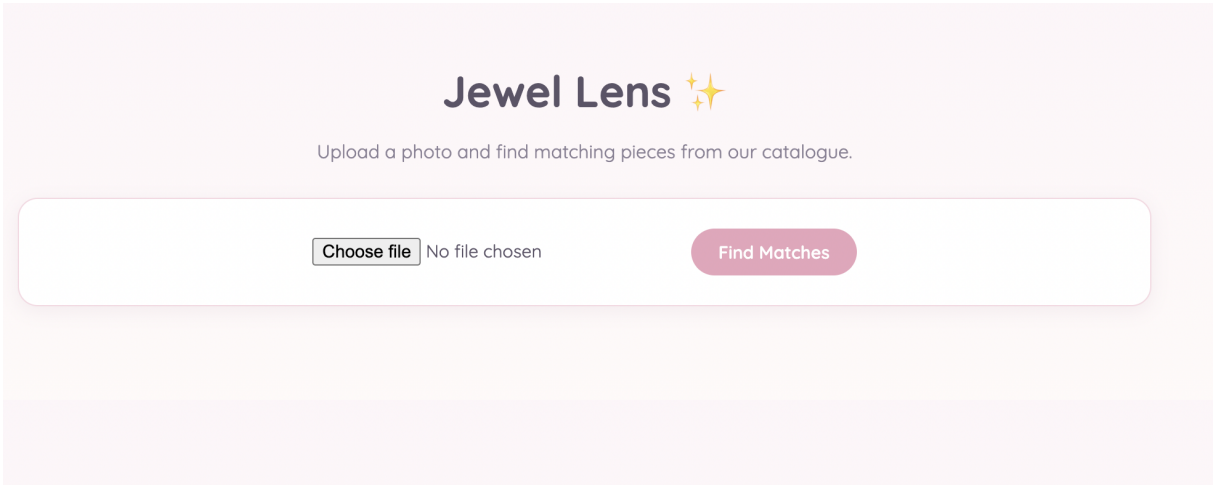
Pipeline: photo → EXIF/resize preprocessing → OWL-ViT zero-shot detection (+ NMS) → per-item crop → DINOv2 frozen embedding → FAISS HNSW similarity search (top-20) + logistic-regression category classifier (soft re-rank boost only) → similarity + margin decision (match / uncertain / no-match) → price/shop metadata attached → top-5 returned, independently per detected item. Full diagram in DECISIONS.md.

No-match handling: a query is only called a match when top-1 similarity clears a high threshold and has a clear margin over the runner-up; otherwise it's uncertain (shown with lower confidence) or no_match. Measured no-match precision on held-out negatives (non-jewelry objects): **0.583** (see Part B results).

Lookup speed: FAISS retrieval itself is near-instant at this catalogue size (single flat HNSW index, no partitioning overhead); detection (OWL-ViT) is the dominant cost per query, not the search step. *[Insert measured per-stage latency numbers here if the profiling script was run — preprocess / detect / embed / retrieve / total.]*

Multi-item extension (the "take it further" item chosen): rather than a separate multi-item model, each detected box is run independently through the full pipeline — this falls out of the architecture for free. Verified via spatial-distinctness + duplicate-result checks: 7/7 multi-item test photos passed cleanly with zero overlap/duplicate warnings.

Frontend

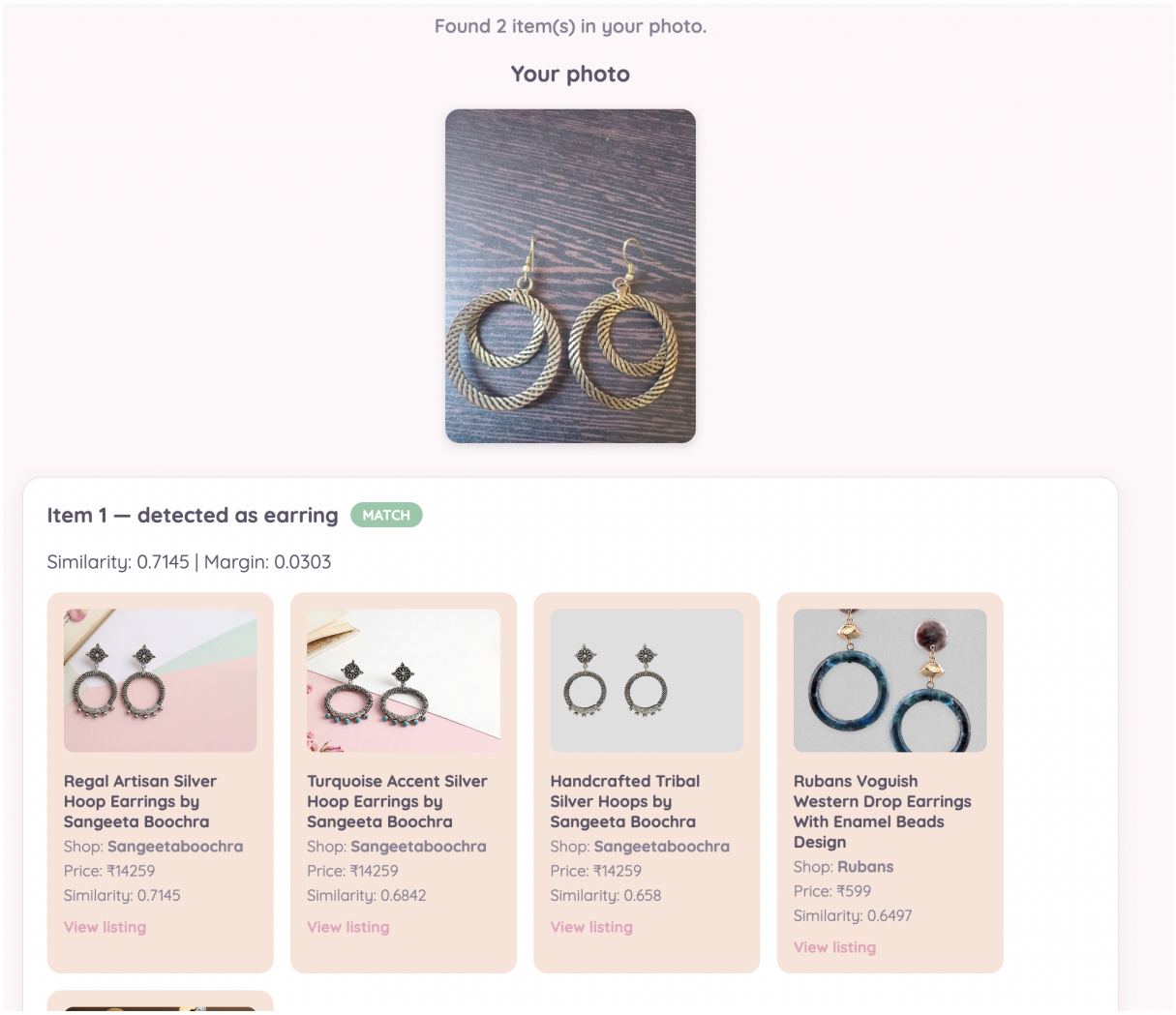


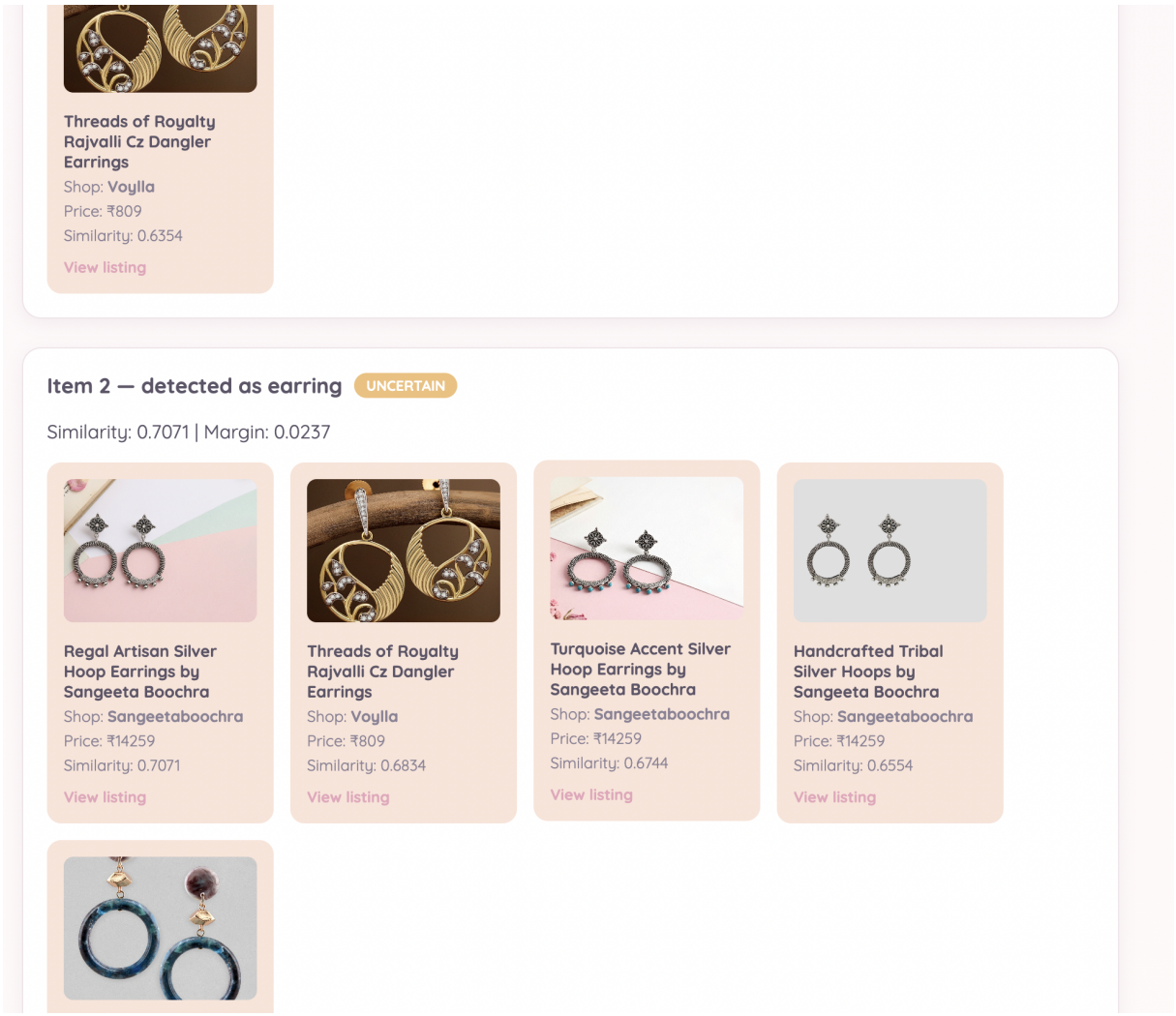
Jewel Lens upload screen

Sample outputs — Part A

Earring

Uploaded photo (pair of hoop earrings) and its detected item results:

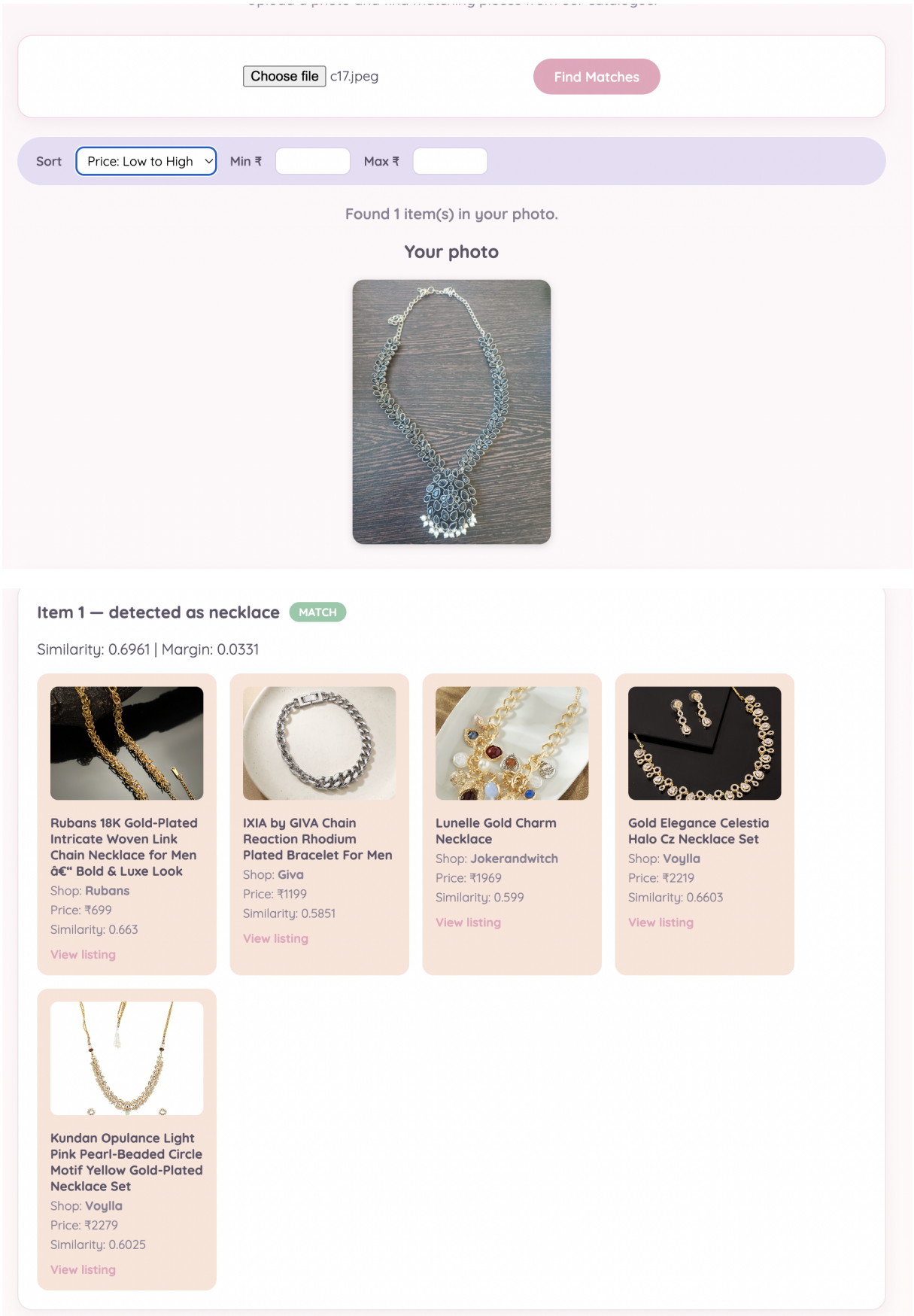




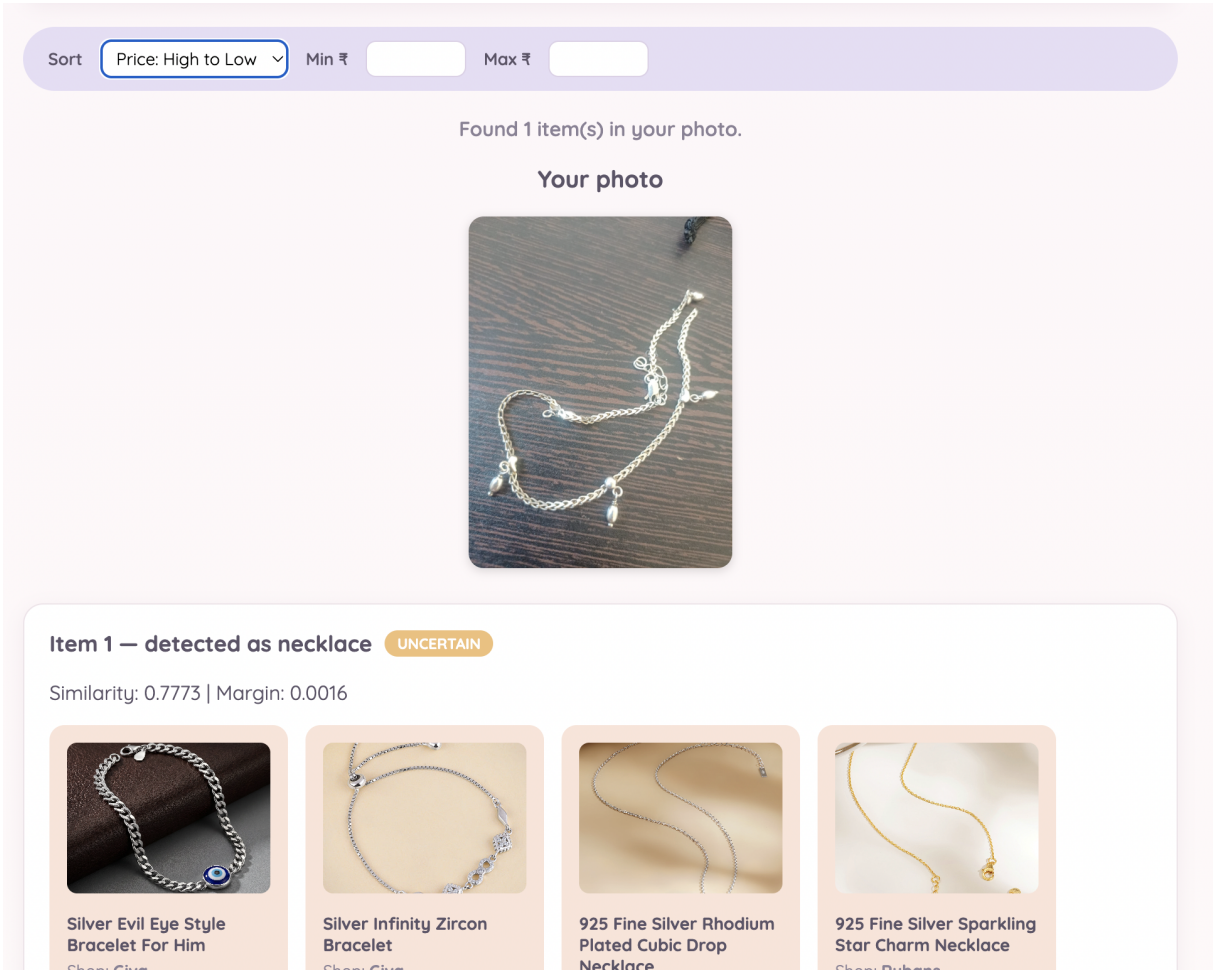
Item 2 result for the same photo

Necklace

Example 1 — uploaded photo and result:

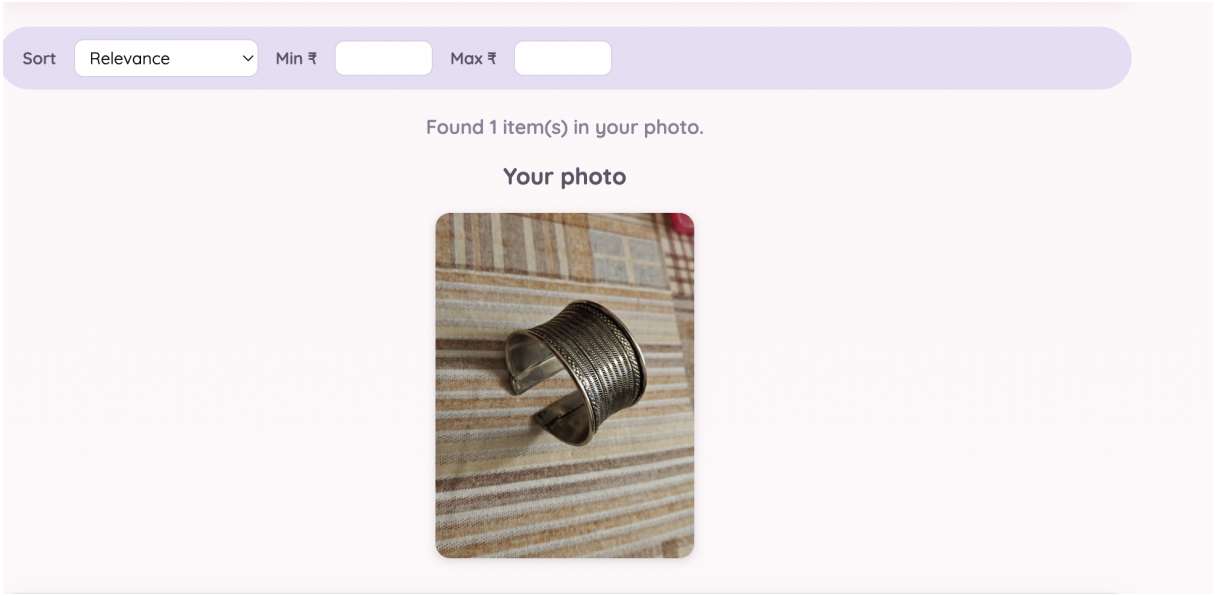


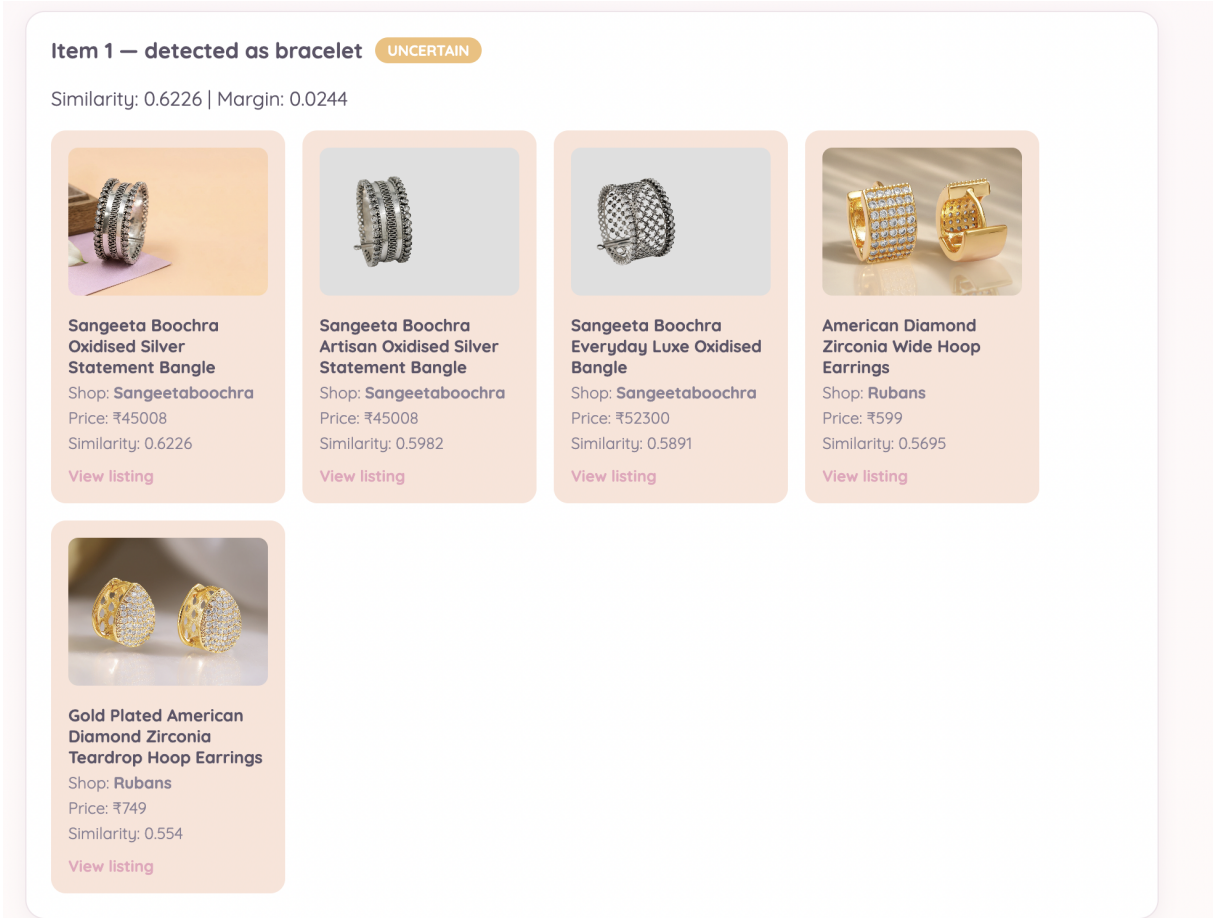
Example 2 — uploaded photo and result:



Bracelet

Uploaded photo (cuff bracelet) and its result:





No-match / negative example

Uploaded photo of a non-jewelry object (toy figurines), correctly rejected with no detections:

Jewel Lens ✨

Upload a photo and find matching pieces from our catalogue.

Choose file dolls.jpeg

Find Matches

No jewelry detected in this photo.

Your photo



Part B — Stumper Test Set

Construction: 100+ self-shot hard photos, organized by folder (earring/, necklace/, bracelet/, mixed/, negatives/), deliberately covering bad lighting, awkward angles, occlusion (hand-held items), blur, reflective surfaces, and multi-item frames. negatives/ holds real non-jewelry objects (brooch, hairclip, keychain, shoes, perfume bottle, etc.) to test rejection.

Harness & metrics

Ground truth from folder structure — independent of the models being tested:

Metric	Result
Detection Recall	1.000 (183/183 jewelry photos got ≥1 detection)
Category Accuracy	0.665
No-match precision (negatives correctly rejected)	0.583
Confused pairs	necklace → earring: 28, bracelet → earring: 27

Exact-match retrieval metrics (Recall@1/5, MRR) were deliberately not reported: the stumper photos are the author's own personal jewelry, never listed on any scraped store, so no catalogue-ID ground truth exists to match against — the "correct" item may legitimately not be in the catalogue at all. Retrieval quality was instead judged qualitatively, by visually reviewing every photo's top-5 results.

Full visual output: review_report.html contains every stumper photo, its detected crop(s), and the 5 candidate matches per item, all self-contained (images embedded as base64). Open it directly in a browser alongside this document — it's the primary evidence for Part B beyond the summary table above.

Error analysis — positives

- Detection boxes are accurate on the large majority of clean, single-item photos.
- Blurred photos were still handled well — DINOv2 embeddings held up reasonably to blur.
- Multi-item photos mostly detect and retrieve each item independently and correctly.
- Even when the upstream category label was wrong, the retrieved images were still visually/stylistically close to the query — retrieval is more robust than either the detector's or classifier's label.

Error analysis — failures (diagnosed root causes, not guesses)

- **Category classifier bias:** dominant confusion is necklace/bracelet → earring, matching a known structural bias in the classifier's learned intercepts (earring is the largest training class; class_weight="balanced" corrects the loss but not the intercepts). Mitigated, not fixed, by keeping the classifier's re-rank influence deliberately small.
- **Round/circular-shaped items (jhumka, hoops) mislabelled as bracelet** — a genuine zero-shot detector limitation: OWL-ViT classifies by 2D silhouette, not object identity, so a closed-loop shape reads as "bracelet" regardless of actual scale or wearing context. The same shape-based confusion also produces pendants

mislabelled as earring (small/dangling silhouette) and hand-held bracelets misclassified as earring once occlusion crops them down to a small curved segment.

- **NMS fails in both directions on the same geometric assumption:** it only suppresses high-IoU box pairs, so (a) a single continuous chain necklace gets split into two boxes (its two ends barely overlap → treated as separate objects), while (b) two genuinely distinct, overlapping worn earrings get merged into one detection (high IoU → treated as duplicates).
- **Mixed-item photos sometimes retrieve a multi-piece "set" listing** for a single detected item — several catalogue products are themselves bundled sets labelled with one category by keyword match; a single-item crop can legitimately have a set-listing as its nearest visual neighbour. This is a catalogue-labelling limitation, not a pipeline bug.

Full trade-off and "with two more weeks" discussion for each of these is in DECISIONS.md.

Repository / running it

See README.md for setup and run instructions (5-minute quickstart with pre-built catalogue artifacts, or full rebuild from scratch).

Architecture & full decision log

See DECISIONS.md for the architecture diagram, every model/architecture decision with alternatives considered, trade-offs made under the time limit, and the complete testing methodology.